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Capivasertib (AZD5363) Summary

Capivasertib is a targeted small-molecule AKT kinase inhibitor used in breast cancer to counteract hyperactivation of the PI3K–AKT–mTOR pathway [1] [2]. It was FDA-approved in 2023 (brand name **TRUQAP®**) for hormone receptor-positive (HR+)/HER2-negative advanced breast cancers harboring activating **PI3K pathway** alterations (PIK3CA mutation, AKT1 mutation, and/or PTEN loss), in combination with fulvestrant after progression on endocrine therapy [2]. In this biomarker-selected population, adding capivasertib to fulvestrant significantly improved progression-free survival (PFS) – reducing risk of progression or death by ~50% versus fulvestrant alone [2]. Major breast cancer guidelines (e.g. NCCN) now list capivasertib+fulvestrant as a **preferred** option for HR+/HER2− metastatic disease with PI3K/AKT pathway alterations, especially after prior CDK4/6 inhibitor therapy [3].

Beyond HR+ disease, capivasertib has been explored across subtypes. Preclinical and early trials showed anti-tumor activity in **triple-negative breast cancer (TNBC)**, particularly in tumors with PI3K/AKT-pathway aberrations [4]. For example, combining capivasertib with paclitaxel modestly improved outcomes in a phase II TNBC trial (PAKT), especially in PIK3CA/AKT1/PTEN-altered TNBC (median PFS 9.3 vs 3.7 months) [5] [6]. However, the larger phase III CAPItello-290 trial in frontline TNBC did **not** meet its primary endpoints: capivasertib+paclitaxel failed to significantly improve overall survival compared to chemotherapy alone [7] [5]. In **HER2-positive** breast cancer, capivasertib has shown preclinical synergy with anti-HER2 therapies (e.g. trastuzumab), since HER2-driven tumors often activate the AKT pathway [4]. Ongoing studies are assessing capivasertib in combination with HER2-targeted agents and other partners. Additionally, capivasertib is being investigated in **BRCA1/2-mutated or HR-deficient** breast cancers in combination with PARP inhibitors (e.g. olaparib), aiming to exploit potential synthetic lethality between AKT inhibition and DNA-repair defects [8]. While results are still preliminary, these efforts highlight capivasertib's broad relevance across molecular subtypes, provided tumors have aberrant PI3K–AKT signaling.

Mechanistically, capivasertib directly antagonizes the **AKT1, AKT2, and AKT3** serine/threonine kinases (also known as PKBα/β/γ) [9], which are central effectors of the PI3K pathway. By ATP-competitively inhibiting all three AKT isoforms with low-nanomolar potency [9] [10], capivasertib blocks downstream signaling that drives cancer cell proliferation, metabolism, and survival. In breast cancer cells, active AKT normally promotes cell-cycle progression and anti-apoptotic signals (e.g. via phosphorylation of substrates like GSK-3β, PRAS40, BAD) and can crosstalk with hormone receptor pathways. Capivasertib's inhibition of AKT causes dephosphorylation/inactivation of these downstream targets, inducing cell-cycle arrest and apoptosis in tumor cells addicted to PI3K–AKT signaling [11] [12]. In HR+ breast cancers, AKT pathway activation is a well-known mediator of endocrine therapy resistance [13]. Capivasertib therefore has a unique role in **endocrine-resistant HR+ tumors**: it can **re-sensitize ER+ cells to hormonal therapy** by disrupting AKT-driven ligand-independent estrogen receptor (ER) activity [14]. For example, AKT inhibition with AZD5363 was shown to restore tamoxifen sensitivity in tamoxifen-resistant cell models and act synergistically with fulvestrant, partly by reducing phosphorylation of mTOR/AKT targets and lowering ERα signaling outputs [14]. In HER2-driven models, capivasertib blocks the HER2→PI3K→AKT cascade, enhancing the efficacy of anti-HER2 agents and suppressing downstream growth signals [4]. Capivasertib may also induce a pro-apoptotic tumor milieu by freeing FOXO tumor suppressors (normally inhibited by AKT) to activate cell-death genes. Overall, **capivasertib's MOA** in breast cancer is to blunt the tumor's key growth/survival pathway (PI3K–AKT) and thereby **halt proliferation, induce apoptosis, and overcome resistance** mechanisms associated with PI3K-pathway hyperactivation (e.g. PTEN loss, PIK3CA mutation, HER2 signaling, or adaptive ER activation) [1] [2].

Identifiers & Synonyms

- **ChEMBL ID:** CHEMBL2325741 [15]
- **DrugBank ID:** DB12218 [16] [17]
- **Synonyms & Codes:** Capivasertib is also known by its research code **AZD5363** (AZD-5363) [18]. The international nonproprietary name (INN) is *Capivasertibum* [19]. It is marketed under the trade name **TRUQAP®** [17]. (No other brand names were identified; "AZC5363" appears as a minor variant code in some databases [15].) All these refer to the same pan-AKT inhibitor.

Mechanism of Action (Breast Cancer Context)

Capivasertib's mechanism of action centers on **pan-AKT inhibition** in human breast cancer. As a selective ATP-competitive inhibitor of AKT1/2/3 [9], capivasertib shuts down the PI3K–AKT–mTOR signaling axis that is aberrantly activated in over half of breast cancers [9]. Key genomic alterations – including **PIK3CA mutations** (~30% of breast tumors), **PTEN loss/mutation** (~35–40%), and **AKT1 activating mutations** (~3–5%) – drive hyperactivation of AKT signaling in both hormone-receptor positive and triple-negative tumors [9] [20]. By inhibiting AKT, capivasertib induces downstream effects highly pertinent to breast cancer biology:

- **Cell Cycle Arrest:** Active AKT normally inactivates FOXO3 transcription factor and stabilizes pro-proliferative factors like **FOXM1**; capivasertib reverses this. In ER+ models, capivasertib requires FOXO3 to exert its effect – relieving AKT's inhibitory phosphorylation of FOXO allows FOXO3 to enter the nucleus and repress cell-cycle drivers [21]. Consequently, capivasertib treatment is associated with **FOXM1 downregulation** and reduced E2F-driven gene expression, halting G_1/S progression [22]. Loss of FOXO3 abrogates capivasertib efficacy, underscoring that **capivasertib works by activating the AKT–FOXO3 tumor suppressor axis**, which in turn dampens proliferation programs [21].

- **Pro-Apoptotic Signaling:** AKT blockade tilts the balance toward apoptosis. Inhibiting AKT prevents phosphorylation of BAD and other pro-apoptotic proteins, promoting mitochondrial apoptosis. It also inhibits mTORC2/PKB feedback, potentially enhancing BIM and other death effectors. Breast cancer cells with PI3K-pathway addictions undergo apoptosis when capivasertib withdraws AKT survival signals [23] [24]. This apoptotic priming is particularly important in tumors that have become **"addicted" to AKT**, such as PTEN-null cancers.

- **Endocrine Modulation:** In ER+ breast cancer, AKT hyperactivation contributes to estrogen-independent growth (through ER phosphorylation and crosstalk). Capivasertib attenuates this crosstalk. Studies showed AKT inhibition by AZD5363 reduces phosphorylated mTOR targets and **diminishes ERα transcriptional activity**, thereby **resensitizing tumors to endocrine therapy** [14]. Clinically, this translates to synergy when capivasertib is combined with fulvestrant: capivasertib blocks the growth-factor signaling that often substitutes for ER signaling in resistant tumors, restoring dependence on the ER pathway which fulvestrant then knocks down [14].

- **Immune and Microenvironment:** While capivasertib is not an immune therapy, PI3K–AKT signaling in tumor cells influences the tumor microenvironment. PTEN loss in tumors can lead to an immune-cold microenvironment; by counteracting PI3K–AKT, capivasertib might indirectly modulate immune infiltration (e.g. increasing T-cell infiltration or altering cytokine profiles), though direct clinical evidence is still emerging. Ongoing studies are examining capivasertib with checkpoint inhibitors for potential immunomodulatory synergy (no definitive human data yet; not cited due to lack of explicit evidence).

In summary, capivasertib **targets a central oncogenic node** in breast cancer. It deprives tumor cells of AKT-mediated signals needed for proliferation (via FOXM1/E2F suppression and cell-cycle arrest) and survival (via apoptosis induction), and it blocks adaptive pathways that confer drug resistance (e.g. feedback ER activation). This makes capivasertib especially effective in contexts where the PI3K–AKT pathway is a key driver of tumor growth or therapy resistance – notably HR+ endocrine-resistant cancers and PI3K-altered subtypes [1] [2].

Primary Targets (Human)

Capivasertib's direct molecular targets in humans (Homo sapiens) are the AKT kinase isoforms:

- **AKT1 (RAC-alpha serine/threonine kinase):** Principal isoform (gene AKT1) often mutated (e.g. AKT1^E17K^) in breast cancer. Capivasertib binds and inhibits AKT1, preventing phosphorylation of downstream effectors like PRAS40, GSK3β, and FOXO3 [9]. In ER+ breast tumors, AKT1 mutations drive estrogen-independent signaling, which capivasertib can counteract [20] [25].

- **AKT2 (RAC-beta serine/thr kinase):** AKT2 (gene AKT2) is another direct target [9]. AKT2 is implicated in insulin/IGF signaling and metabolic control. In some breast cancers (especially endocrine-resistant and metastatic disease), AKT2 supports survival and motility. Capivasertib's inhibition of AKT2 contributes to its blockade of the PI3K–AKT axis in cancer cells [9].

- **AKT3 (RAC-gamma serine/thr kinase):** AKT3 (gene AKT3) is also directly inhibited [9]. AKT3 is expressed in some breast cancers (e.g. triple-negative subtype, brain metastases) and can mediate growth signals particularly in HER2+ or basal-like tumors. Capivasertib's pan-AKT activity (inhibiting AKT1/2/3 with similar potency) ensures that no AKT isoform "escapes" inhibition [9].

Binding affinity: Capivasertib has low-nanomolar IC_50s (≈3–8 nM) against each AKT isoform [10]. It does **not** significantly target upstream kinases (PI3K, PDK1) or other AGC kinases at therapeutic concentrations, although at higher exposures some off-target effects (e.g. against PKA or RSK) have been noted preclinically. The primary intended targets, however, are AKT1-3, as confirmed by biochemical assays and cellular phosphorylation readouts [10] [26]. By inhibiting AKT, capivasertib functionally impacts multiple pathways (cell cycle, apoptosis, metabolism), but these effects stem from AKT blockade rather than direct inhibition of those downstream molecules.

Pathways (Overview)

Capivasertib perturbs several key pathways in breast cancer, both as part of its mechanism of action and through tumor adaptive responses. Below is an overview of **10 major pathways** influenced by capivasertib, along with their formal identifiers and context:

- **PI3K/AKT/mTOR Signaling (HALLMARK_PI3K_AKT_MTOR_SIGNALING):** The primary pathway targeted by capivasertib. Upstream oncogenic events (PIK3CA mutation, PTEN loss, HER2 signaling) activate this hallmark pathway in breast cancer [9] [20]. Capivasertib inhibits AKT, thereby blocking downstream mTORC1 activity, protein synthesis, and cell growth signals. *Context:* Mechanistic target; pathway hyperactivation correlates with drug sensitivity (tumors "addicted" to PI3K–AKT are more susceptible) [27] [1].

- **FOXO–FOXM1 Transcriptional Axis (Reactome:** FOXO-mediated transcription**, part of cell cycle regulation):** Capivasertib triggers a shift in this axis. AKT inhibition frees **FOXO3** to enter the nucleus and downregulate pro-proliferative genes like **FOXM1** [21]. *Context:* In ER+ PIK3CA-mutant breast cancers, the AKT–FOXO3–FOXM1 axis determines response – drug efficacy requires FOXO3-driven FOXM1 suppression [22]. Upregulation of FOXO activity (and thus cell-cycle checkpoint genes) is associated with sensitivity, whereas persistent FOXM1 signaling confers resistance.

- **Estrogen Receptor Signaling (Gene Ontology: GO_0038146 – ER signaling pathway):** This hormone-driven pathway is indirectly affected. AKT can phosphorylate ER or its co-regulators, sustaining ER-driven transcription in endocrine-resistant tumors. Capivasertib's blockade of AKT leads to reduced ER phosphorylation and activity [14]. *Context:* In HR+ cancers, **downregulating ER signaling** (e.g. with fulvestrant) alongside AKT inhibition produces synergistic anti-tumor effects [14]. Thus, the ER pathway, when suppressed, becomes a **sensitivity-associated** axis for capivasertib (especially in Luminal A/B subtypes).

- **HER2/RTK Signaling (Reactome: R-HSA-1227960 "Signaling by ERBB2"):** Many HER2+ breast cancers activate PI3K–AKT through HER2/HER3. Capivasertib blocks the downstream AKT node. However, tumor cells often invoke **bypass RTK signaling** as feedback – e.g. increased HER3 or EGFR activity when AKT is inhibited. *Context:* Upregulation of RTK/MAPK pathways can mediate resistance (adaptive reactivation of HER/EGFR–RAS–ERK signaling when AKT is off) [24]. Combining capivasertib with HER2 inhibitors or MAPK pathway inhibitors is a strategy to prevent this compensation.

- **RAS/MAPK Pathway (KEGG: hsa04010 "MAPK signaling pathway"):** Capivasertib doesn't directly target MAPK, but there is **crosstalk**. In response to AKT inhibition, tumor cells can exhibit feedback **ERK1/2 hyper-activation** [24]. This occurs because relieving PI3K–AKT feedback loops (e.g. via mTORC1/S6K) upregulates receptor signaling, funneling into RAS–ERK. *Context:* **ERK upregulation** is a known **resistance mechanism** to AKT inhibitors – high phospho-ERK can drive proliferation despite AKT blockade [24]. Conversely, dual AKT+MEK/ERK inhibition yields more potent anti-tumor responses, underscoring that the MAPK pathway can compensate when upregulated.

- **mTORC1 Protein Translation (KEGG: hsa04150 "mTOR signaling pathway" / Hallmark_MTORC1_SIGNALING):** AKT is a key activator of mTORC1; capivasertib should reduce mTORC1 activity and cap-dependent translation. Indeed, responders to capivasertib show suppressed phosphorylation of S6 ribosomal protein and 4EBP1 (mTORC1 substrates). However, if tumors manage to maintain high mTORC1-driven translation, they tend to resist the drug [28] [29]. *Context:* **Elevated mTORC1 output** (e.g. high levels of translation initiation proteins) is linked to capivasertib resistance, even in PI3K-mutant tumors [28]. This pathway is thus both a target (to be downregulated for efficacy) and a marker of resistance if inadequately inhibited.

- **DNA Damage Response (DDR) / Homologous Recombination (Hallmark_DNA_REPAIR, Reactome: R-HSA-5693538 "Homologous Recombination"):** There is interplay between AKT and DNA repair. AKT can promote HR repair (e.g. via phosphorylation of DNA-PKcs and BRCA1). Capivasertib may impair efficient DNA repair, leading to accumulation of DNA damage in tumor cells. *Context:* Tumors with **pre-existing HR defects** (e.g. **BRCA1/2-mutated**) could be extra sensitive to AKT inhibition, especially in combination with PARP inhibitors [30]. While not yet clinically proven as a single-agent biomarker, this pathway is being explored – early trials of capivasertib + olaparib showed promising activity in HR-deficient cancers, hinting that downregulation of HR repair can increase capivasertib sensitivity (synthetic lethality) (evidence still investigational – no robust human data to cite for single-agent capivasertib).

- **Apoptosis and BCL-2 Family Regulation (GO:0008625 "Extrinsic apoptotic signaling" & GO: 0008285 "negative regulation of apoptosis"):** AKT directly phosphorylates pro-apoptotic factors (BAD, Caspase-9) to prevent apoptosis. Capivasertib lifts these brakes, promoting apoptotic cell death. Tumors with intact apoptotic machinery (e.g. functional BAX/BAK, low anti-apoptotic BCL2) respond better once AKT is inhibited. *Context:* **Upregulation of apoptotic pathway activity** (e.g. BIM induction via FOXO3) correlates with drug sensitivity, whereas tumors that **downregulate apoptosis** (via p53 loss or BCL2 overexpression) show resistance. (For instance, cells lacking BIM or p53 may resist capivasertib-induced death – though specific clinical proof in breast cancer is limited, it aligns with general apoptosis biology.)

- **EMT and Invasion (Hallmark_EPITHELIAL_MESENCHYMAL_TRANSITION):** The PI3K–AKT pathway influences EMT; PTEN loss/Akt activation can drive mesenchymal transition and metastasis. Capivasertib may impede EMT by inhibiting AKT-driven transcription factors (TWIST, SNAIL) that promote EMT. In some models, knocking down AKT or SGK1 reduced invasion and migration [31]. *Context:* Mesenchymal/EMT-high tumors are often more resistant to therapies. There is evidence that high SGK1 (associated with mesenchymal phenotype) confers AKT inhibitor resistance [32] [33]. Thus, **EMT-related pathway upregulation (TGFβ/SGK1)** can be a resistance hallmark, whereas maintaining an epithelial phenotype (E-cadherin high, SGK1 low) might favor sensitivity.

- **Immune Microenvironment and Cytokine Signaling (ImmuneSigDB: e.g. "TNFA_SIGNALING_VIA_NFKB"):** AKT in tumor cells modulates production of cytokines (like VEGF, IL-6) and PD-L1 expression. Capivasertib might reduce immunosuppressive cytokines and PD-L1 levels (AKT/mTOR upregulates PD-L1), potentially enhancing anti-tumor immunity. In TNBC, AKT inhibition has been linked to increased T-cell infiltration in preclinical models (not yet confirmed clinically). *Context:* This is a relatively minor pathway in terms of direct evidence, but ongoing trials combining capivasertib with immunotherapy suggest that changes in **immune signaling pathways** (e.g. interferon gamma response) are being monitored. No definitive pathway call-outs from clinical data yet, so immune pathways remain exploratory in the context of capivasertib.

These pathways underscore that capivasertib's impact in breast cancer is multifaceted. Its primary effect is on the PI3K–AKT–mTOR pathway (mechanistic link), but numerous **sensitivity** and **resistance** pathways modulate outcomes – from cell cycle and ER signaling (especially in HR+ subtypes) to parallel signaling (RTK/MAPK) and adaptive survival circuits (mTORC1 translation, SGK1, EMT). In the detailed sections below, we delve into which pathways are upregulated or downregulated in various contexts and how they relate to drug sensitivity or resistance.

Upregulated Pathways (by Context/Subtype)

This section covers pathways found to be transcriptionally or functionally **upregulated in the context of capivasertib treatment or resistance**, stratified by breast cancer subtype and scenario. Upregulated pathways can either indicate an adaptive resistance mechanism (tumors turning on alternative survival routes when AKT is inhibited) or, in some cases, reflect biomarkers of sensitivity (pathways that are high at baseline, making the tumor more dependent on AKT).

1. Adaptive feedback upregulation in resistant tumors: One consistent finding is that inhibition of AKT triggers compensatory **MAPK pathway upregulation**. In multiple models (including breast cancer cell lines and patient samples), capivasertib treatment led to a **feedback increase in ERK1/2 phosphorylation** [24]. For example, a study in solid tumor models showed that capivasertib (and other AKT/PI3K inhibitors) caused **hyperactivation of ERK** in all tested cell lines due to relief of negative feedback on receptor tyrosine kinases [24]. This **MEK–ERK feedback loop** is especially relevant in PTEN-deficient or PIK3CA-mutant cancers – when AKT is turned off, HER family receptors (EGFR, HER3) and other upstream signals ramp up, activating RAS–RAF–MEK–ERK. **Subtype context:** This upregulation occurs across subtypes (HR+ and TNBC alike) as a general resistance mechanism. Clinically, tumors that exhibited high MAPK pathway activity on-treatment tended to be less responsive. Thus, **upregulation of the RAS/MAPK pathway** is a hallmark of adaptive resistance to capivasertib. Combination approaches (AKT + MEK inhibition) have been proposed to counter this feedback [24].

2. FOXM1 and cell-cycle gene upregulation in resistance: Long-term capivasertib exposure in ER+ breast cancer cells was found to **upregulate the FOXM1-driven transcriptional program** as an escape mechanism [22]. In capivasertib-resistant cell lines derived from ER+ PIK3CA-mutant models, **FOXM1 mRNA and protein were markedly elevated** relative to sensitive cells [22]. FOXM1 is a transcription factor governing G_2/M and mitotic genes; its increase suggests the cells are bypassing AKT/FOXO-mediated cell-cycle arrest. *Subtype:* This was shown in Luminal (ER+) context [34]. Notably, **FOXM1 upregulation** correlated with loss of FOXO3 activity (since FOXO3 normally represses FOXM1). Therefore, a high-FOXM1 state denotes a pathway upregulation that **drives resistance** (continued proliferation despite AKT inhibition). In contrast, responders showed FOXM1 downregulation. This adaptive upregulation of FOXM1 (and associated E2F targets) is a key resistant-pathway in ER+ cancers.

3. SGK1 and alternative AKT-substitute upregulation: Serum/glucocorticoid-regulated kinase 1 (SGK1) is a PI3K-pathway effector paralogous to AKT, controlled by the same upstream signals (PI3K, PDK1, mTORC2). Research demonstrated that many breast cancer cells inherently resistant to AKT inhibitors have **elevated SGK1 activity** [32]. Specifically, resistant cell lines displayed high SGK1 expression and phosphorylation of its substrate NDRG1, whereas sensitive lines had low SGK1 [32]. Although this is baseline upregulation (not necessarily caused by capivasertib, but rather a parallel pathway being up), it functionally means that in the presence of capivasertib, tumor cells **up-regulate reliance on SGK1** to drive survival signals. In experiments, AKT-inhibitor-resistant cells could be growth-inhibited by knocking down SGK1, confirming that SGK1 upregulation substitutes for AKT [35]. *Subtype:* SGK1 upregulation has been observed mainly in triple-negative or basal-like contexts (which often have PTEN loss). Thus, **high SGK1 signaling** is an upregulated pathway in certain tumors that correlates with resistance to capivasertib.

4. mTORC1 and protein synthesis upregulation: Some tumors manage to keep the mTORC1–translation pathway active despite AKT inhibition. Proteomic analysis of PIK3CA-mutant breast tumors treated with capivasertib monotherapy found that non-responding tumors had **higher baseline and on-treatment levels of proteins reflecting active mTORC1-driven translation** (e.g. ribosomal proteins, elongation factors) [28] [29]. In other words, an **upregulated protein synthesis pathway** (often via mTORC1/eIF4E) was a feature of resistance. Mechanistically, these tumors might have alternative inputs to mTORC1 (such as RAS or nutrient-sensing pathways) that remain active, or partial inhibition of AKT allows mTORC1 reactivation. *Subtype:* This was seen in a mix of HR+ cases with PIK3CA mutations [36] [28]. The key point is that **persistent mTORC1 activity** (high translation) is an upregulated hallmark in resistant tumors.

5. Immune checkpoint and inflammatory pathway upregulation (context-specific): There is some evidence that tumors might compensate for AKT blockade by modulating immune pathways. For instance, AKT inhibition can lead to increased **STAT3 activation** in PTEN-null models as a feedback (via IL-6 cytokine loops) [37] [38]. STAT3 upregulation could promote survival and an immunosuppressive environment. While not extensively documented in clinical breast tumors, one could see **inflammatory pathway upregulation** (IL-6/STAT3, NF-κB) as an escape. Additionally, **PD-L1 expression** can be induced by MEK/ERK activity when AKT is suppressed (since AKT/mTOR inhibition sometimes triggers a PI3K–Jak–STAT feedback). *Subtype:* likely in basal/TNBC which are more immune-infiltrated. This remains a theoretical consideration; explicit evidence in patients is limited, so it's noted as a potential upregulated pathway in resistance rather than a confirmed one.

In summary, **upregulated pathways in response to capivasertib** include those that provide **bypass survival signals** or allow the cell cycle to continue: notably, the RAS–MAPK pathway, the FOXM1/cell-cycle axis, alternative AGC kinases like SGK1, and persistent mTORC1 signaling. These upregulated signals are more often associated with **resistance** (tumors that don't respond or that acquire resistance show these high-pathway activities). Importantly, these findings are human-derived (cell lines, patient tumors) and underscore why combination therapies are being explored (e.g. adding MEK inhibitors to tackle ERK feedback, or mTOR inhibitors to ensure translation is shut down, etc.).

Downregulated Pathways (by Context/Subtype)

This section highlights pathways that are found to be **downregulated** either as a result of capivasertib exposure or in certain sensitive contexts. Pathway downregulation can indicate mechanisms through which capivasertib exerts its anti-tumor effect (i.e. pathways that the drug suppresses), and in some cases, loss of a pathway might paradoxically drive resistance (if the pathway was a necessary component of cell death).

1. Cell-cycle progression and E2F targets – downregulated in responders: In breast tumors that respond to capivasertib, there is often a prominent **downregulation of cell-cycle gene signatures**. Specifically, effective AKT inhibition in ER+ cancers led to **downregulation of the E2F/FOXM1 transcriptional program** [39]. Treated tumors show decreased expression of genes controlling G_1–S and G_2–M (cyclins, CDKs, mitotic kinesins, etc.), consistent with cell-cycle arrest. FOXM1, as mentioned, is strongly downregulated in capivasertib-sensitive models [39]. This pathway suppression is largely mediated by FOXO3 reactivation – once AKT is off, FOXO3 (a cell-cycle brake) represses these genes. *Subtype:* Most evident in HR+ (Luminal) cancers with PIK3CA mutations [34], but cell-cycle downregulation is a universal marker of proliferation arrest, so TNBCs that respond may similarly show Ki-67 and E2F target drops. Thus, **"Cell cycle/mitotic pathways" down** is a hallmark of sensitivity.

2. Estrogen receptor signaling – downregulated by combination therapy: While capivasertib alone doesn't directly degrade ER, in the context of combined therapy, **ER signaling pathways are effectively downregulated**. In endocrine-resistant models, capivasertib reduced expression of ER-dependent genes and in combination with ER downregulators led to a broader shutdown of ER signaling [14]. For example, genes in the **estrogen response early/late** Hallmark sets were suppressed when AKT was inhibited (since AKT can phosphorylate and activate ER or its coactivators, its inhibition attenuates that). Clinically in CAPItello-291, pairing capivasertib with fulvestrant caused deeper blockade of the ER pathway than fulvestrant alone (evidenced by more complete inhibition of ER-regulated proliferation). *Subtype:* HR+/HER2− tumors – this is the rationale behind the combo; **downregulation of ER pathway** by fulvestrant + capivasertib correlates with tumor regressions [14]. If an ER+ tumor fails to downregulate ER signaling (e.g. ESR1 mutation driving constitutive activity), that could mediate resistance.

3. AKT/mTOR downstream signaling – intended downregulation: By mechanism, capivasertib should **downregulate AKT downstream pathways**: notably **mTORC1 signaling and protein translation**. In responders, one observes decreased phosphorylation of S6 and 4E-BP1 (mTORC1 outputs), indicating lowered translation initiation [28]. Also, **glycolytic and biosynthetic pathways** driven by AKT/mTOR get dampened – e.g. glucose uptake genes, lipid synthesis genes are downregulated as the tumor shifts to a slower metabolic state. *Subtype:* Across all; this is a direct pharmacodynamic effect. However, incomplete downregulation of mTORC1 (as noted, if 4E-BP1 phosphorylation remains high) is a bad sign. Thus, we consider "**Protein synthesis/translational pathway downregulation**" as a feature of effective treatment; lack thereof is resistance.

4. DNA repair and HR pathway – potential downregulation (synthetic lethality context): AKT activity intersects with the DNA damage response. In some contexts, AKT inhibition can **downregulate expression or function of BRCA1/2 and RAD51** (via FOXO or other mechanisms), thereby exacerbating homologous recombination deficiencies [40] [41]. If a tumor is HR-deficient, capivasertib might further cripple compensatory repair, leading to accumulation of DNA damage and cell death. *Subtype:* BRCA1/2-mutant or "HRD+" breast cancers (can be TNBC or HR+). For instance, preclinical work suggests **AKT inhibition is synthetic lethal with BRCA1 loss**, as it impairs PCNA ubiquitination and TLS DNA repair [40] [41]. So, although not a classical "transcriptional downregulation" measured in patients, we include **homologous recombination repair pathway down** as a sensitizing context (lack of HR is a vulnerability that capivasertib can exploit in combination with DNA-damaging agents). Evidence in human breast tumors is still emerging, so this is noted with caution.

5. Pro-survival feedback signals – downregulated in combination strategies: Tumors often activate survival pathways (like IGF-1/Insulin signaling) in response to drugs. Combining capivasertib with other inhibitors can lead to **downregulation of those feedback loops**. For example, adding capivasertib to an mTOR inhibitor might further **downregulate insulin receptor signaling** (since AKT is a node for insulin signaling). Or combining with a MEK inhibitor could **downregulate both PI3K and MAPK survival signals**, leaving no major pathway for feedback. In trials, capivasertib + everolimus (mTORi) or capivasertib + PD-1 inhibitors have shown that many compensatory cytokine/growth-factor signals (like IL-6, heregulin) are reduced in presence of dual blockade (no definitive publication yet, so consider this an extrapolation from known biology).

In summary, **downregulated pathways associated with capivasertib** include those that define a successful therapeutic response: cell-cycle/E2F suppression, ER signaling shutdown (in HR+ combos), metabolic slowdown via mTORC1 inhibition, and potentially a collapse of the tumor's DNA repair capacity in HRD contexts. These pathway decreases are generally linked to **increased drug sensitivity** – they are goals of therapy. If certain pathways fail to be downregulated (e.g. if FOXM1 remains high or translation remains active), that often signals resistance (addressed above). Importantly, no "paradoxical" pathway downregulation causing resistance has been firmly identified in human data – i.e. there isn't a case where suppressing a pathway leads the tumor to resist (except perhaps if a pro-apoptotic pathway is lost, see Resistance Pathways for Downregulation below).

Sensitivity Pathways for Upregulation (by Context/Subtype)

"Sensitivity pathways for upregulation" refers to pathways whose **higher activity or expression correlates with *increased* sensitivity** to capivasertib in breast cancer. In other words, when these pathways are upregulated (either inherently in the tumor or induced), the tumor becomes more susceptible to capivasertib's effects. We identify the following key pathways where upregulation is associated with **greater drug sensitivity**, along with subtype context:

- **PI3K–AKT Pathway Hyperactivation:** Perhaps counterintuitively, the very pathway capivasertib targets – **PI3K/AKT signaling** – when strongly upregulated, confers sensitivity to the drug. Tumors that are "addicted" to AKT signaling are more likely to respond. Clinically, breast cancers with activating alterations in this pathway (e.g. **PIK3CA mutations, PTEN loss, AKT1^E17K^ mutation**) derive the most benefit from capivasertib [25] [42]. In the phase III trial, patients with tumors harboring PIK3CA/AKT1/PTEN alterations (which cause upregulated PI3K–AKT activity) had significantly improved PFS on capivasertib [42]. *Subtype:* This spans multiple subtypes: HR+ tumors often have PIK3CA mutations; some TNBC have PTEN loss. For example, AKT1^E17K^-mutant ER+ breast cancers showed a 40% response rate to capivasertib monotherapy [43], demonstrating that **upregulation of the AKT pathway (via mutation)** makes the tumor highly sensitive to AKT blockade. **Pathway ID:** *KEGG hsa04151 (PI3K-Akt signaling pathway)*. **Rationale:** The tumor becomes overly reliant on AKT for survival; blocking it causes collapse [1] [2].

- **FOXO3/Pro-apoptotic Signaling:** Upregulation (activation) of the **FOXO3 pathway** enhances sensitivity. As mentioned, capivasertib works partly by engaging FOXO3. Tumors with intact and inducible FOXO3-mediated transcription undergo apoptosis and cell-cycle arrest upon AKT inhibition. If a tumor cell can strongly turn on FOXO3 target genes (like BIM, p27^Kip1^) when AKT is inhibited, it will be more sensitive. In the ER+ context, **high FOXO3 activity** (sometimes inferred from gene expression profiles) predicted better response [21]. Conversely, FOXO3 deletion (loss of that pathway) caused resistance [22]. *Subtype:* Shown in ER+ PIK3CA-mutant models [27]. Thus, **upregulation of the FOXO-driven apoptotic program** is a sensitivity mechanism. This can be viewed as "HALLMARK_APOPTOSIS" up – cells primed for apoptosis (e.g. low BCL2, high pro-apoptotic factors) are killed more readily by capivasertib. **Pathway ID:** *KEGG hsa04068 (FoxO signaling pathway)*.

- **Loss of Negative Regulators (PTEN) – effectively upregulating AKT:** When a negative regulator is down, the pathway it restrains is upregulated. A prime example is **PTEN loss**. PTEN normally keeps PI3K signaling in check; its loss leads to unchecked AKT activation. Tumors with PTEN loss (which is an "upregulation" of PI3K pathway activity) are especially dependent on AKT, and capivasertib shows efficacy there [27] [22]. Notably, PTEN-null/low HR+ cancers benefited from capivasertib in trials, whereas they derive less benefit from upstream PI3K inhibitors [27] [25]. *Subtype:* HR+ (5–10% have PTEN homozygous deletion) and TNBC (PTEN loss in ~35%). **Pathway ID:** *REACTOME:R-HSA-8943724 (PTEN Regulation of PI3K/AKT pathway)*. **Rationale:** Upregulation of PI3K signaling via PTEN loss creates a dependency that capivasertib exploits.

- **HER2–PI3K Pathway Activation:** In HER2+ breast cancers, high HER2 signaling drives PI3K–AKT. Tumors with **HER2 pathway upregulation** (e.g. overexpression of HER2/heregulin) might be particularly sensitive to AKT inhibition when combined with HER2 blockade. Preclinical studies showed capivasertib plus anti-HER2 therapy was most effective in cell lines with **upregulated HER2→AKT signaling** [4]. *Subtype:* HER2-enriched. If a HER2+ tumor has high PI3K activity (perhaps due to PIK3CA mutation as well), it stands to gain from AKT inhibition. **Pathway ID:** *Reactome R-HSA-1250347 (PI3K events in ERBB2 signaling)*.

- **mTORC2/SGK-dependent pathways (context: dependence on AKT paralogs):** Some basal-like cancers have upregulation of PDK1/mTORC2 signaling that feeds both AKT and SGK. If a tumor upregulates a pathway that still **requires AKT** (for example, some metabolic circuits require AKT even if SGK is present), those tumors can be sensitive – but this is a nuanced scenario. In general, **tumors that have not upregulated significant AKT-independent pathways remain sensitive**. So one could say, paradoxically, *lack* of upregulation of bypass pathways = sensitivity. But focusing on positive: **Upregulation of any pathway that increases reliance on PI3K–AKT** (like high insulin/IGF signaling driving AKT) makes the tumor more capivasertib-sensitive because blocking AKT cuts off that key output. For instance, if a tumor has an autocrine IGF2 loop (IGF2 upregulation) activating AKT, it will be highly sensitive to AKT inhibition (this is reported in some endocrine-resistant cases).

- **Immune cell presence (T-cell inflammation):** Though data are limited, there's a hypothesis that tumors with an **"immune-hot" microenvironment** might be more sensitive to AKT inhibition. AKT in tumors can promote immunosuppressive factors; thus, an immune-active tumor may benefit if AKT inhibition tips the balance further toward immune clearance. In one trial, TNBC patients with immune gene signatures seemed to do well with AKT inhibitor + chemo (though hard to isolate AKT effect) [44]. We list this as speculative: **upregulation of interferon-γ and cytotoxic T-lymphocyte pathways** could synergize with the cell-intrinsic effects of capivasertib. *Subtype:* TNBC (especially those that are PD-L1 positive). Not a primary determinant, but a potential contributing sensitivity factor when combined with immunotherapy.

In summary, **pathways whose upregulation increases capivasertib sensitivity** are essentially those that make the cancer **highly dependent on AKT signaling**. Key among these: the PI3K–AKT pathway itself (genetic activation), the downstream pro-apoptotic execution (FOXO3 activation), and certain context-specific axes like HER2. High pathway activity = high dependency, which capivasertib can exploit, leading to synthetic lethality or dramatic responses [42] [43].

Sensitivity Pathways for Downregulation (by Context/Subtype)

This category describes pathways where **reduced activity (downregulation) enhances the efficacy** of capivasertib. In other words, if a particular pathway is suppressed, the tumor becomes more susceptible to

AKT inhibition. These often represent pathways that would otherwise compensate for or counteract the drug's effect – when they are knocked down or absent, capivasertib works better. Key examples:

- **FOXM1 & Cell-Cycle Pathway Downregulation:** As discussed earlier, FOXM1 is a proliferation-driving TF. **Downregulation of FOXM1** (through genetic or pharmacologic means) markedly increases sensitivity to capivasertib [39] . In resistant ER+ cells, experimentally knocking down FOXM1 restored their response to capivasertib [39] . This implies that if the cell-cycle/mitotic drive is blunted, AKT inhibition more effectively induces arrest and death. *Subtype:* ER+ (where FOXM1 is often elevated in endocrine resistance). Clinically, there's no direct FOXM1 inhibitor yet, but FOXM1-low tumors tend to do better. **Pathway ID:** *Hallmark E2F_TARGETS / G2M_CHECKPOINT*. **Rationale:** Without FOXM1, cells cannot bypass the cell cycle blockade induced by capivasertib, so they succumb.

- **SGK1 Pathway Downregulation:** Tumors with low or suppressed **SGK1** are more sensitive to capivasertib. As shown, SGK1-high conferred resistance; conversely, SGK1 knockdown selectively impaired growth of AKT inhibitor–resistant cells, essentially re-sensitizing them [35] . This indicates that if the **SGK1 axis is downregulated**, capivasertib has an easier time killing the cell (no alternate route). *Subtype:* PTEN-null TNBCs with inherently low SGK1 expression responded better to AKT inhibition than those with SGK1 induction. In practical terms, a tumor lacking an SGK1-mediated escape is reliant solely on AKT – a vulnerability. **Pathway ID:** Possibly *GO:1905938 (regulation of SGK signaling)*. **Rationale:** SGK1 down = one less backup for AKT, so AKT inhibitor hits with full force.

- **MAPK/ERK Pathway Downregulation:** If the RAS–ERK pathway is quiet, capivasertib is more effective. Many combination studies show that when you pharmacologically downregulate MEK/ERK (with MEK inhibitors), the sensitivity to AKT inhibitors improves, suggesting the reverse is also true: a tumor with inherently low MAPK activity won't have strong compensatory survival upon AKT blockade. *Subtype:* Some HR+ tumors have minimal RTK/RAS activity and are more purely PI3K-driven; these tend to respond well to capivasertib (because they lack the MAPK escape route). In contrast, tumors that were both PI3K- and RAS-driven needed dual inhibition. **Pathway ID:** *KEGG hsa04010 (MAPK pathway)*. **Rationale:** Downregulating the MAPK pathway removes a major resistance mechanism, thereby increasing capivasertib efficacy [24] .

- **Estrogen/ER Pathway Downregulation:** This is exemplified by combining endocrine therapy with capivasertib. **Downregulation of the ER signaling axis** (via SERDs like fulvestrant or aromatase inhibition lowering estrogen) significantly enhances capivasertib's effect in ER+ cancers [14] . Essentially, dual suppression of ER and AKT is far more lethal to Luminal tumors than either alone. So one can phrase: in ER+ context, **ER pathway downregulation is a sensitivity pathway** – it "clears the way" for AKT inhibitor to fully execute cell death without ER-driven survival signals. *Subtype:* HR+/HER2−. This was proven in clinical trials (capivasertib+fulvestrant outperformed capivasertib alone, and certainly fulvestrant alone) [2] . **Pathway ID:** *HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE*. **Rationale:** ER signaling up can rescue cells via alternative growth programs; its downregulation synergizes with AKT inhibition to induce tumor regression.

- **DNA Repair (HR) Deficiency:** If a tumor has **downregulated HR repair** (e.g. BRCA1 mutation or epigenetic BRCA1 silencing), it may be more susceptible to capivasertib, especially in combination with DNA-damaging therapy. Preclinical data suggest AKT inhibition in HR-deficient cells leads to catastrophic DNA damage accumulation and cell death (AKT normally helps facilitate DNA repair via RAD51; its inhibition in HRD cells is devastating) [40] [41] . For example, AKT inhibitor + PARP inhibitor showed synthetic lethality in BRCA-mutant models. We cautiously include this: in trials, patients with BRCA1/2 mutations experienced tumor shrinkage on capivasertib + olaparib, indicating a particular vulnerability [30]. Thus, **HR pathway downregulation (HRD)** is a sensitizing context. *Subtype:* gBRCA1/2-mutated breast cancers (often TNBC or HR+/HER2− with HRD). **Pathway ID:** *Reactome R-HSA-983231 (DNA Double-Strand Break Repair via HR)*.

- **mTORC1/Protein Synthesis Downregulation:** Tumors that naturally have lower mTORC1 activity or protein synthesis rates might be more easily pushed over the edge by AKT inhibition (they have less buffering capacity). In one analysis, responders had lower baseline phospho-4EBP1 (indicating less mTORC1 drive) [45]. When capivasertib was given, these tumors completely shut off translation and underwent apoptosis. Conversely, tumors with very high baseline protein synthesis (e.g. due to 4EBP1 loss) were harder to fully inhibit. So inherently **downregulated mTORC1 signaling** (or effective pharmacologic downregulation via combos like capivasertib + everolimus) correlates with sensitivity. *Subtype:* no specific, more of a tumor phenotype.

In essence, **pathways where less is more (for sensitivity)** are those that otherwise would undermine capivasertib's effect. By **downregulating compensatory pathways (FOXM1, SGK1, ERK, ER, etc.)**, either inherently or with combination therapy, we remove survival crutches the tumor might use, thereby **potentiating capivasertib's efficacy**. Evidence for these comes from both mechanistic studies (e.g. knockdowns reversing resistance) and clinical logic (e.g. combining therapies to force multiple pathway downregulation) [39] [35].

Resistance Pathways for Upregulation (by Context/Subtype)

"Resistance pathways for upregulation" are those whose **increased activity leads to resistance or reduced response** to capivasertib. When these pathways are upregulated (either pre-existing in a tumor or induced during treatment), they allow the cancer to bypass or counteract capivasertib's action. Several key upregulated pathways associated with resistance have been identified:

- **MAPK/ERK Pathway Upregulation:** As noted earlier, a major resistance mechanism is **reactivation of the MAPK cascade**. In tumors where **ERK phosphorylation/activity goes up** during capivasertib therapy, outcomes are worse [24]. This upregulation can occur via various routes: increased expression of receptor tyrosine kinases (like EGFR, HER3, FGFR), RAS mutations (less common in breast ca), or relief of feedback loops. The net effect is that the cancer shifts its dependency from AKT to the RAS–ERK pathway. Clinically, this was observed in biomarker studies where high baseline Ras/MAPK signatures were associated with primary resistance to AKT inhibitors [46] [24]. *Subtype:* This spans subtypes but is especially noted in PTEN-null TNBC (which often activate EGFR–MAPK when AKT is blocked) and some HER2+ tumors (which have redundant signaling via MAPK). **Example:** A lung cancer study (analogous mechanism) explicitly showed capivasertib increased ERK phosphorylation as compensatory feedback [24]; in breast cancer, combining AKT + MEK inhibitor was synergistic, implying ERK upregulation was limiting capivasertib efficacy [47]. **Thus, upregulation of the MAPK pathway** is a clear driver of resistance.

- **FOXM1/E2F Pathway Upregulation:** Tumors that **upregulate FOXM1 and associated cell-cycle genes** in spite of AKT inhibition tend to be resistant. For instance, in ER+ models that acquired resistance, FOXM1 levels were dramatically elevated (the opposite of what happens in sensitive cells) [22]. Overexpression of FOXM1 experimentally **reduced capivasertib efficacy** [39]. Mechanistically, high FOXM1 means the cell cycle continues (through G_2/M) despite AKT/FOXO3 signals. *Subtype:* ER+ (especially after long-term drug exposure) [21]. Patients whose tumors exhibit high proliferation gene signatures (e.g. high Ki-67, E2F targets) on-treatment are less likely to respond – essentially the tumor did not shut down its replication machinery. **Upregulation of the FOXM1-driven transcriptional program** is therefore a resistance pathway, allowing the tumor to bypass AKT-induced cell-cycle arrest.

- **SGK1/Alternative AGC Kinase Upregulation:** High **SGK1** is a known mediator of resistance to AKT inhibitors [32] [35]. Upregulation of SGK1 (and possibly related kinases like PKA or PKC in some contexts) provides an alternate means to phosphorylate certain downstream targets of AKT. Sommer *et al.* (2013) showed that breast cancer cells with **elevated SGK1** were intrinsically resistant to both capivasertib and the allosteric AKT inhibitor MK-2206 [32]. SGK1 can maintain phosphorylation of substrates like NDRG1 and GSK3β even when AKT is off, partially substituting for AKT function. Clinically, tumors with high SGK1 mRNA/protein might be less responsive. *Subtype:* This is particularly relevant in PTEN-deficient, PI3K-activated basal-type cancers – some of these upregulate SGK1 via mTORC2 signaling. **Thus, upregulation of the SGK1 pathway** (or broadly, the PI3K–PDK1–SGK axis) is a route to resistance. Combination of AKT and SGK inhibitors is being considered as a way to address this [33].

- **mTORC1 Reactivation/Protein Synthesis:** A dominant resistance driver found in a proteomic study was **reactivation of mTORC1-dependent translation** despite capivasertib [28] [29]. Tumors that resisted capivasertib had higher levels of ribosomal and translation proteins (e.g. eIF4E, phosphorylated 4EBP1). Essentially, they managed to keep the protein synthesis machinery upregulated. This could be via AKT-independent mTORC1 activation (maybe through nutrient-sensing pathways or loss of 4EBP1). *Subtype:* Observed in HR+ PIK3CA-mutant cases that derived no benefit from capivasertib [45]. The authors concluded **increased mTORC1-driven translation functions as a mechanism of resistance** [28]. Upregulation of this pathway means the drug fails to induce the metabolic/catabolic crisis it normally would. It's an "intrinsic" resistance if present at baseline and could be adaptive if the tumor finds a way to reactivate mTOR while on drug.

- **Upregulation of Compensatory PI3K Nodes (e.g. PIM kinases, p70S6K):** Tumors might upregulate other kinases that lie parallel or downstream. For example, **PIM kinases** (PIM1/2) are oncogenic serine-thr kinases that can compensate for AKT in some contexts (they can phosphorylate some of the same substrates). If a tumor upregulates PIM2, it could continue to drive mTOR and Myc despite AKT inhibition. This has been noted as a possible resistance route in other cancers; in breast, it's speculative but plausible. Similarly, **upregulation of MEK5/ERK5 pathway** or others could provide alternate proliferation cues.

- **EMT and MET Pathways:** Upregulation of EMT transcription factors (like ZEB1, SNAIL) and mesenchymal phenotype often correlates with multi-drug resistance, including to AKT inhibitors. If capivasertib-treated cells undergo an EMT (which can be induced by TGF-β pathway upregulation), they may become less proliferative (hence less drug-sensitive in terms of needing AKT) and more reliant on alternative pathways. One aspect of EMT is activation of focal adhesion kinase (FAK) signaling. Some data suggests that the feedback activation of ERK upon AKT inhibition is **attenuated by FAK inhibition** [48], meaning cell adhesion signals (integrins/FAK) play a role. So if a tumor upregulates integrin/FAK signaling (common in EMT/high-motility cells), it might reduce dependency on AKT and confer resistance. *Subtype:* Basal/claudin-low TNBC often have EMT traits and might be less responsive to AKT inhibitors unless combined with other agents.

In summary, **pathways whose increased activity yields resistance** form a picture of the tumor finding alternate routes to survive and proliferate when AKT is blocked. High MAPK signaling, high cell-cycle drive (FOXM1/E2F), activation of kinases like SGK1, and resurgence of mTORC1 activity are all **hallmarks of an upregulated resistance state** [32] [28]. These upregulated pathways either bypass the AKT blockade or neutralize its effects, underscoring the need for combination therapies to preempt or overcome such resistance.

Resistance Pathways for Downregulation (by Context/Subtype)

Finally, “resistance pathways for downregulation” refers to pathways where a **decrease or loss of activity leads to resistance**. At first glance, one might assume removing a pathway would always help the drug (less activity to overcome), but in some cases the absence of a certain pathway makes the drug less effective. These often involve **loss of pro-sensitivity factors or loss of pathways that the drug relies on to kill the cell**. Key examples in capivasertib resistance include:

- **FOXO3 Pathway Downregulation:** Loss or downregulation of **FOXO3** is a prime example of this category. As noted, FOXO3 is critical for executing the effects of AKT inhibition (inducing cell-cycle arrest and apoptosis). If a resistant tumor manages to **downregulate or inactivate FOXO3** (e.g. through deletion, mutation, or continued AKT-independent phosphorylation), capivasertib’s efficacy plummets [21]. In the 2025 npj Breast Cancer study, deletion of FOXO3A in ER+ cells rendered them significantly less responsive to capivasertib [21]. Essentially, the pathway of FOXO-mediated transcriptional repression was turned off, so the drug could not trigger the anti-proliferative program. *Subtype:* ER+ PIK3CA-mutant contexts (where FOXO3A loss of function could emerge as a resistance mechanism) [39]. Clinically, tumors with low FOXO3 (due to, say, co-mutations in FOXO3 or upstream constant IκB (which holds FOXO out of nucleus)) may exhibit primary resistance. **Thus, downregulation of the FOXO3 pathway** (no activation of FOXO target genes) leads to capivasertib resistance.

- **Pro-apoptotic Pathway Downregulation:** If a tumor has **intrinsic defects in apoptosis execution**, it can resist capivasertib. For example, **downregulation of BAX/BAK or upregulation of anti-apoptotics** (BCL-2, MCL1) can be viewed as the apoptotic pathway being suppressed. In practical terms, if a breast cancer has lost p53 function (common in TNBC) or has low pro-apoptotic factor expression, it may not undergo apoptosis even if AKT is inhibited. While not a “pathway downregulation” in terms of gene set, it is a functional pathway loss: the apoptotic machinery is muted. *Subtype:* p53-mutant TNBCs might require combination with pro-apoptotic drugs for capivasertib to work. Though not always framed as “resistance to AKT inhibitor” in literature, it fits: **downregulation of the apoptosis pathway** (through loss of key effectors) confers the ability to survive AKT blockade, a resistant phenotype. Supporting evidence: some AKT inhibitor studies in other cancers note that BIM deletion or BCL2 overexpression (i.e. turning off the apoptotic response) yields resistance; similar principles likely apply in breast cancer (e.g. tumors with high MCL1 could resist until that is addressed).

- **Estrogen Receptor Downregulation in ER-dependent context:** Interestingly, if an HR+ tumor **loses ER expression** (becomes independent of estrogen), capivasertib (especially given with endocrine therapy) might have diminished effect. In the context of combination, the presence of ER was part of the dual targeting; an ER-negative (or very low ER) tumor might rely purely on growth factor pathways that adapt quickly. For instance, if an initially ER+ tumor undergoes lineage switch or shuts down ER (e.g. via ESR1 methylation or fulvestrant doing its job too well but the tumor cells survive), those cells might activate other pathways (like MAPK, as above) and be less susceptible to AKT inhibition. *Subtype:* This could occur in Luminal tumors that undergo endocrine treatment–related evolution. There isn't direct evidence that "ER loss causes AKT inhibitor resistance," but conceptually, **downregulation of the ER pathway** removes one vulnerability that capivasertib+fulvestrant was exploiting (the dual hit). We mention this as a potential mechanism: in trials, patients who had completely endocrine-insensitive disease (no ER-driven component) needed that capivasertib had to do all the work, which if not sufficient, leads to resistance. (Emerging data suggests capivasertib still has some effect even in ESR1-negative cases, but likely less pronounced.)

- **Downregulation of PTEN reactivating PI3K** – a nuanced case: PTEN loss we listed under sensitivity (because it upregulates AKT). Conversely, **if a tumor somehow restores PTEN or downregulates PI3K signaling**, it might become less sensitive to AKT inhibitor (simply because AKT pathway is no longer a key driver). This is more a case of loss of dependency: e.g. if through some evolution a tumor activates MAPK and in doing so downregulates PI3K activity (feedback loops), it might not care about AKT anymore – thus resistant. There's no clear clinical scenario where PTEN comes back, but tumor heterogeneity might present PTEN-wildtype clones that are less dependent on AKT and thus resistant. So one can consider **downregulation of the very pathway target (AKT signaling)** in parts of the tumor as a mechanism of resistance – those cells wouldn't be killed by capivasertib. However, most often resistance is via upregulation of compensators rather than actual downregulation of PI3K, so this is a minor point.

- **Immune surveillance pathway downregulation:** If capivasertib's effect partially relies on immune-mediated killing (still hypothetical), then a tumor that downregulates MHC I or antigen presentation might resist immunogenic cell death. But this is speculative and not documented.

In summary, **pathways whose suppression leads to resistance** are generally those that are needed for capivasertib to induce its tumor-killing effect. Chief among them is the **FOXO3/tumor-suppressor pathway** – losing this means capivasertib cannot execute transcriptional changes required for cell death [21]. Another is the **apoptotic cascade** – if it's blunted, the drug's pro-death signal doesn't translate to cell kill. Unlike the "upregulated resistance pathways" which are more about alternate survival routes, the "downregulated resistance pathways" are about **loss of vulnerability** – the tumor sheds the very feature the drug was exploiting. These are important to note because they highlight that not all resistance is about finding a new path; some is about removing the target's context (e.g. losing FOXO3 or changing lineage). If evidence for these is insufficient or inferred (like the ER loss scenario), we clearly indicate so. For FOXO3, however, the evidence is concrete: *FOXO3A deletion = capivasertib resistance* [21].

Breast Cancer Subtype–Stratified Evidence

Breast cancer is heterogeneous, and capivasertib's efficacy and mechanisms can vary by subtype. Here we summarize how the above findings apply to specific subtypes:

- **HR+/HER2– (Luminal A/B):** This is the setting where capivasertib has shown clear clinical benefit (CAPItello-291). In HR+ tumors, PI3K pathway activation (PIK3CA mutant, AKT1 mutant, PTEN loss) is common and correlates with endocrine resistance [49]. Capivasertib is most effective in Luminal tumors with those alterations, as it targets the escape pathway from hormonal control [2]. Mechanistically, HR+ tumors rely on both ER and AKT signaling; capivasertib restores endocrine sensitivity by blocking AKT-driven ER phosphorylation and cell-cycle progression [14]. **Subtype-specific evidence:** Turner *et al.* 2023 showed capivasertib+fulvestrant improved PFS both in PI3K-pathway-altered HR+ tumors (median PFS 7.2 vs 3.6 months) and even to a lesser extent in pathway-wildtype HR+ tumors [42], suggesting broad utility but stronger when alterations are present. Resistance in HR+ often involves upregulation of downstream proliferative signals (FOXM1) or alternate RTKs (like FGFR) after prolonged endocrine/AKT suppression [22]. Also, ESR1 mutations (which keep ER active) don't prevent capivasertib benefit, but complete loss of ER (lineage change) would remove the combination's synergy. HR+ patients need monitoring of glucose/lipid levels on capivasertib (due to metabolic effects) but otherwise tolerate it along with endocrine therapy [8]. In summary, for Luminal subtypes: **PI3K/AKT-driven luminals respond well; key resistance nodes are cell-cycle upregulation and SGK1 (in PTEN-null cases).**

- **HER2+:** No approved role yet, but preclinical data suggest synergy. Many HER2+ tumors have PIK3CA mutations (~20-30% of HER2+ cases) and those tend to do worse on HER2-targeted therapy alone. Capivasertib can inhibit the AKT downstream of HER2 and potentially overcome resistance to anti-HER2 agents. For instance, capivasertib + trastuzumab had preclinical efficacy in cell lines, particularly when **PI3K–AKT was hyperactive** [4]. Ongoing trials (e.g. CAPItello-291 included a subset of HER2-low, which is basically HR+ with low HER2; separate studies are looking at HER2+). We anticipate that in HER2+ cancers, **resistance pathways upregulated** might include parallel MAPK (since HER2 signals both PI3K and MAPK). If a HER2+ tumor has an active AKT1 mutation (rare but possible) or PTEN loss, it could be an ideal candidate for capivasertib. Conversely, if a HER2+ tumor is driven more by RAS/ERK or has lost dependence on AKT (e.g. through PIK3CA Y1021 mutation that affects binding), capivasertib may be less effective. So, while not standard yet, **there is a rationale to target AKT in HER2+ disease with aberrant PI3K signaling**, and trials are exploring this (no level 1 evidence yet to cite beyond preclinical).

- **Triple-Negative (TNBC):** TNBC often has PI3K pathway dysregulation (PTEN loss, PIK3CA mutations in maybe 15% of TNBC, AKT3 upregulation). Early-phase trials hinted at benefit: the PAKT trial (phase II) in metastatic TNBC found that capivasertib + paclitaxel significantly improved OS, especially in the subset with PIK3CA/AKT1/PTEN alterations [5] [6]. However, the larger phase III CAPItello-290 failed to show an OS benefit in an unselected TNBC population [7]. This suggests that **biomarker selection is crucial in TNBC**. In TNBC patients whose tumors had PI3K-pathway mutations, PFS was much longer with capivasertib (9.3 vs 3.7 months as mentioned) [6], echoing that these "addicted" tumors are sensitive. But TNBC tumors without such alterations likely activate compensatory pathways (and indeed TNBC is quick to engage MAPK, SGK, or immune evasion routes). Resistance in TNBC was often associated with **feedback EGFR/HER3 and MAPK upregulation**, as well as potentially greater reliance on glycolysis (some TNBC lines increase GLUT1 and go glycolytic to survive AKT inhibition). SGK1-mediated resistance was first discovered in TNBC cells (MDA-MB-468, BT-549 were among those studied) [32] [35]. *Subtype specifics:* Basal-like TNBC might require dual PI3K/AKT and MAPK targeting; mesenchymal TNBC might not respond well unless combined with other therapies. If TNBC has BRCA mutation, capivasertib + PARP inhibitor is a strategy being tested (CAPItello-292 includes some TNBC). So, **for TNBC, capivasertib shows promise when a clear PI3K–AKT driver is present; otherwise, TNBC quickly upregulates bypass pathways leading to modest results**.

- **gBRCA-mutated / HRD+:** This overlaps with TNBC often, but also some HR+ have BRCA mutations. Preclinically, BRCA1-deficient cells were sensitive to AKT inhibitors, especially in combination with DNA damage. Clinically, there isn't a lot of direct data for capivasertib monotherapy in BRCA-mutated breast cancer. However, an ongoing trial (FAKTION) looked at capivasertib+fulvestrant and noted that even BRCA-mutated patients benefited similarly if they had PI3K pathway activation – no negative interaction there. The concept is that **HRD tumors might not repair the damage exacerbated by AKT pathway suppression** (AKT supports HR via BRCA1/2), so they could be more sensitive. In other words, if a BRCA1-mutant TNBC also has a PI3K mutation, capivasertib could be doubly effective. *One caution:* HRD tumors often activate compensatory ATR/CHK1 pathways; combining capivasertib with ATR inhibitors is another synthetic lethal approach being investigated (outside scope). In sum, **BRCA/HRD status is not a formal requirement for capivasertib, but it could enhance the drug's impact or guide combinations (like adding PARPi)** – evidence still limited.

- **Basal-like vs Luminal vs HER2-enriched (Intrinsic subtypes):** Basal-like (many TNBC) tumors frequently have EGFR or AKT3 activation; capivasertib might need help (chemo or EGFR inhibitor) in those. Luminal B (high proliferation HR+) often harbor PI3K mutations – ideal for capivasertib + endocrine. Luminal A (lower grade HR+) may not have as frequent PI3K mutations, but if present and if the patient progressed on CDK4/6, capivasertib is an option. HER2-enriched (whether clinically HER2+ or not) tend to have high AKT activity as well, so they conceptually fit, but data needed. **Triple-positive** (ER+HER2+) cancers: combination with endocrine and HER2-targeted plus capivasertib could be a future exploration.

In conclusion, **capivasertib's utility is clearest in HR+/HER2− breast cancer with PI3K/AKT pathway aberrations** (now an approved indication) [2]. It has activity in TNBC, but patient selection is critical – ongoing research is defining which TNBC subgroups benefit (likely those with PI3K pathway mutations or AKT1 E17K). Each subtype brings its own dominant pathways, and resistance mechanisms often align with those: ER+ tumors resist via cell-cycle/FOXM1 or FGFR; TNBC resist via MAPK/SGK or EMT; HER2+ might resist via parallel signaling. Recognizing these subtype differences is vital for designing effective combination therapies (for example, adding FGFR inhibitors in ER+ FGFR-amplified cases, MEK inhibitors in TNBC with MAPK feedback, etc.).

(Evidence note: The above stratification is drawn from multiple sources – pivotal trial data for HR+ [2], subset analyses for TNBC [5] [6], and mechanistic studies for ER+ resistance [22] and TNBC SGK1 resistance [32]. If a particular subtype mechanism is not strongly evidenced in humans (e.g. immune aspects), it is presented as a hypothesis with the available rationale.)

Contraindications and Safety

Capivasertib's contraindications and safety profile have been characterized in regulatory approvals and trials:

- **Contraindications:** The only absolute contraindication listed is **severe hypersensitivity to capivasertib or any of its components** [50]. Patients with a history of anaphylaxis or severe allergic reaction to capivasertib should not receive it. There are no known **molecular contraindications** (e.g. it's not contraindicated based on tumor biomarkers – if anything, lack of PI3K mutations means less benefit but not a safety contraindication). Notably, **pregnancy** is contraindicated due to potential harm to the fetus (capivasertib, like many anti-cancer agents, can cause embryo-fetal toxicity). While the official label language advises pregnancy avoidance rather than listing it under "Contraindications," practically pregnant patients should not be treated with capivasertib. Similarly, breastfeeding is advised against during and for a period after treatment.

- **Diabetes/Hyperglycemia Precautions:** Capivasertib can induce significant hyperglycemia (due to AKT's role in insulin signaling) [51] [52]. In the CAPItello-291 trial, 37% of patients on capivasertib experienced elevated fasting glucose, and about 11% had ≥Grade 2 hyperglycemia [53]. **Patients with Type 1 diabetes or uncontrolled Type 2 diabetes were excluded from trials** [54], and the label notes that safety in those populations is not established [54]. So, while diabetes is not an absolute contraindication, it is a **relative contraindication/precaution** – type 1 diabetics or those requiring insulin should use capivasertib only with extreme caution and close monitoring, if at all [54]. In practice, one would optimize blood glucose and perhaps avoid capivasertib if diabetes is brittle.

- **Other Warnings:** There are important **warnings** (not strict contraindications) for:

- **Severe hyperglycemia** including rare cases of diabetic ketoacidosis [51] – patients need monitoring of blood glucose frequently, especially in the first weeks [55]. If persistent Grade 3 hyperglycemia occurs despite medication, treatment may be withheld or discontinued [56].
- **Dermatologic reactions:** Capivasertib can cause rash (including maculopapular rash, acneiform rash). In trials, Grade 3 rash occurred in ~12-16% [8]. Severe cutaneous reactions (e.g. Stevens-Johnson Syndrome) haven't been prominent, but the most common adverse event leading to discontinuation was "cutaneous adverse reactions" (in 6% of patients) [57]. Patients with a history of severe skin reactions to PI3K/AKT inhibitors might be at risk.
- **Diarrhea:** Very common (any grade ~60%, Grade ≥3 in ~14-17% [58]). Monitor and manage aggressively with antidiarrheals and dose modifications if needed.
- **Hepatic effects:** Some elevation in ALT/AST was seen (Grade 3 ALT ~2.6%) [59]. No formal contraindication in hepatic impairment, but caution: patients with severe hepatic impairment (Child-Pugh C) were not well-studied. Capivasertib is metabolized (partly via CYP3A), so severe liver dysfunction could increase drug levels. The label likely recommends caution or dose adjustment in moderate hepatic impairment (to be checked in full PI).
- **Renal effects:** AKT inhibitors can cause AKI in rare cases by volume depletion (due to diarrhea) or hyperglycemia osmotic diuresis. No contraindication, but patients with severe renal impairment weren't included in some studies.
- **Interactions:** Capivasertib is primarily metabolized by CYP3A4; coadministration with strong CYP3A4 inducers or inhibitors is not contraindicated but requires dose adjustments. Also, drugs that themselves cause hyperglycemia (like steroids) could exacerbate this AE.

- **Cardiac:** Unlike some PI3K inhibitors, capivasertib hasn't shown strong QT prolongation. However, AKT pathway can affect cardiac metabolism; ongoing monitoring of any arrhythmias or heart failure symptoms is prudent, especially if patient has underlying heart disease.

- **Population-specific:**

- Geriatric use: No specific contraindication; older patients tolerated it similarly, though careful monitoring of AEs (older patients might be more prone to dehydration from diarrhea or to hyperglycemia).

- Pediatric: Not studied, no role in pediatrics (breast cancer is an adult disease).
- **Fertility:** As with many anti-cancer drugs, fertility could be affected. Contraception is advised for women of childbearing potential during and after treatment (per label, probably for at least a month or two after last dose).
- **Contraindications per guidelines:** NCCN and others don't list special contraindications beyond the above (they focus on appropriate patient selection). NCCN does mention not to give capivasertib to a patient with uncontrolled diabetes until it's controlled, due to hyperglycemia risk.

In summary, **capivasertib is contraindicated in patients with severe hypersensitivity to it** [50] **.** It should be avoided (or used only with extreme caution) in pregnancy and in patients with uncontrolled diabetes mellitus. There are no tumor biology–based contraindications, but it's indicated only in certain biomarker-positive cancers (using it outside that context isn't "contraindicated" per se, just not evidence-supported). The main safety concerns that must be managed are **hyperglycemia**, **rash**, and **diarrhea** [58] [51]. Supportive care (e.g. metformin or insulin for hyperglycemia, loperamide for diarrhea, prophylactic antihistamines/creams for rash) can allow most patients to continue therapy. In trials, dose interruptions/reductions occurred in about 20–25% of patients to manage side effects [60], and 10% discontinued due to AEs (most commonly rash) [61].

No "black box warning" exists for capivasertib, but the FDA label emphasizes monitoring blood glucose and skin toxicities [51] [53]. As a novel targeted agent, long-term safety is still being observed. However, from current data, **capivasertib's safety profile is considered manageable** with appropriate monitoring, and contraindications are limited mainly to hypersensitivity (and pregnancy by standard practice) [50].

Trial and Guideline Context

Capivasertib's development in breast cancer has been guided by clinical trials and is now reflected in treatment guidelines:

- **Key Clinical Trials:**
- *FAKTION (Phase II):* An early trial in endocrine-resistant HR+ MBC where capivasertib + fulvestrant showed a PFS improvement over placebo + fulvestrant (median 10.3 vs 4.8 months, HR 0.54) [62]. This provided proof of concept that adding AKT inhibition can overcome endocrine therapy resistance in ER+ tumors, especially in those with PI3K pathway aberrations.
- *CAPItello-291 (Phase III, NCT04305496):* This pivotal trial confirmed the benefit in a larger population. It enrolled HR+/HER2– advanced breast cancer patients who had progressed on aromatase inhibitor + CDK4/6 inhibitor. Patients were stratified by PI3K pathway mutation status. Capivasertib + fulvestrant significantly improved median PFS (7.2 vs 3.6 mo in all patients, HR ~0.60) [2] and overall response rate (ORR ~23% vs 12%). Notably, benefit was seen in **both the biomarker-altered group** (PIK3CA/AKT1/PTEN, where PFS ~7.3 vs 3.1 mo, HR ~0.50) **and the biomarker-wildtype group** (PFS ~6.4 vs 3.7 mo, HR ~0.70) [42], although it was greater in the altered subset. OS data are immature but trending favorably. These results led to approvals. This trial also collected translational samples; analyses like the proteomic study (Sobsey et al. 2024) came from CAPItello-291 participants, identifying the mTORC1 translation resistance marker [45].
- *CAPItello-290 (Phase III, NCT03997123):* Evaluated capivasertib + weekly paclitaxel in untreated metastatic TNBC (n=~800). It **did not meet its primary endpoints** of improving PFS/OS overall or in the biomarker-positive subset as co-primary endpoints [7]. Detailed results: interim analysis reported no significant OS improvement (median OS ~19 mo both arms) [63], even though earlier

phase II (PAKT) had indicated OS benefit. PFS difference was modest. It suggests that in unselected TNBC, adding AKT inhibitor to chemo may not be sufficient, possibly due to rapid emergence of resistance (as discussed). Subgroup data hinted that patients with PI3K-pathway-altered TNBC might have had better outcomes, but the trial wasn't formally positive. This was a setback for the TNBC indication – hence capivasertib is not broadly used there, except perhaps in clinical trials or off-label for a known AKT1-mutant TNBC (e.g. AKT1^E17K^, which some oncologists might try given the dramatic responses seen in some case series [43]).

- *CAPItello-292 (Phase III, NCT04416718):* Ongoing trial combining capivasertib with **palbociclib + fulvestrant** in HR+ MBC. This targets an earlier line (first-line endocrine-resistant setting) to see if triple therapy (CDK4/6 + AKT + endocrine) further improves outcomes or delays resistance compared to current standard (palbo+fulvestrant). No results yet reported as of late 2025.

- Other trials: *IPATunity130* (using ipatasertib, a different AKT inhibitor) in a similar HR+ setting was stopped early for futility in the unselected population, but did show some benefit in an AKT1-mutant subgroup. Capivasertib appears more broadly effective than ipatasertib, possibly due to patient selection and drug properties. Trials in combination with **PARP inhibitors** (e.g. NCT04060862 for capivasertib+olaparib in various solid tumors including BRCA-mutant breast) have shown tolerability and some efficacy, supporting the synthetic lethality rationale. Also, a neoadjuvant trial I-SPY 2 is evaluating if capivasertib can increase pathologic complete response rates when added to chemo in certain biomarker-defined breast cancers.

- **Guideline Recommendations:**

- The **NCCN Guidelines (2025)** list capivasertib + fulvestrant as a Category 1 recommended option for **HR+/HER2– metastatic breast cancer with a PI3K/AKT pathway alteration**, post one line of endocrine therapy (typically after CDK4/6 inhibitor) [3] . They specify that at least one of PIK3CA, AKT1, or PTEN must be mutated/altered (based on FDA approval) [3] . The NCCN notes to confirm the mutation via an FDA-approved test (e.g. next-gen sequencing or PCR for PIK3CA). For patients without such alterations, NCCN currently does not recommend capivasertib (since the trial showed less pronounced benefit, and other options like everolimus or chemotherapy exist). However, clinicians might consider it if no other options, given some activity even in wildtype.
- **ESMO** and other international guidelines likely align: ESMO (2023) in their ABC guidelines mentioned that AKT inhibitors (capivasertib) are an emerging option especially for PI3K-pathway mutant HR+ cancers, likely to be incorporated once approved. As of 2024, ESMO would say after progression on CDK4/6 + AI, if PIK3CA-mutant then alpelisib+fulvestrant is preferred; if PIK3CA-wild but has AKT1/PTEN mutation, capivasertib+fulvestrant is an option (or even if PIK3CA-mutant and one prefers capivasertib due to better tolerance than alpelisib's rash/hyperglycemia profile).
- **ASCO** has not yet issued separate guidelines specifically for capivasertib, but in general, expert opinion echoes the above: test for PI3K pathway mutations; if present, consider adding capivasertib after CDK4/6 resistance.
- **Regional differences:** In the US, FDA approval (Nov 2023) covers PIK3CA/AKT1/PTEN-altered cases [2] . In the EU, EMA approval (2024) is similar; NICE in the UK is likely evaluating cost-effectiveness. A Drug cost vs benefit analysis (JMCP 2023) suggested capivasertib is cost-effective as a second-line therapy for those mutations [64] .
- For **TNBC:** No guideline endorses capivasertib for TNBC outside trials. If a TNBC patient has an AKT1 E17K mutation (rare ~1-2% of TNBC), a clinician might seek off-label use or trial, given a JAMA Network study showed some responses in various AKT1-mutant tumors including breast [43] . But formally, chemo and immunotherapy remain standard for TNBC.

- **HER2+:** Not in guidelines for HER2+; standard anti-HER2 therapies prevail. Possibly mention in future if trials like AKT in refractory HER2+ succeed.

- **Ongoing Research and Future Directions:** Multiple trials are combining capivasertib with other agents:

- **CAPItello-292** (mentioned) will inform if upfront triple therapy yields enough benefit to change practice (some worry about toxicity triple combination).
- Trials with **immune checkpoint inhibitors** in TNBC (since AKT inhibition might modulate tumor immunogenicity and because the negative CAPItello-290 left an opening to see if adding immunotherapy could help).
- **Neoadjuvant trials:** to see if short-term capivasertib can induce tumor shrinkage and pathway changes pre-surgery (particularly in PI3K-mutant HR+ where you might add it to endocrine therapy to increase complete cell kill).
- The concept of **sequential therapy:** If a patient has a PIK3CA mutation and progresses on alpelisib (a PI3Kα inhibitor), could they still benefit from capivasertib? Possibly, since capivasertib covers more alterations and downstream. Some case reports show responses to AKT inhibitor after PI3K inhibitor, especially if the mechanism of resistance was say an AKT mutation that bypassed PI3Kα inhibition. This isn't in guidelines yet, but something the oncology community is noting.

In summary, **capivasertib has secured a role in the HR+ metastatic setting** per trials and guidelines [2] [3], and its use is guided by molecular profiling. Trials in other settings (TNBC, HER2+) have been less conclusive so far. Guidelines emphasize proper patient selection, proactive management of side effects (with input from endocrinologists for hyperglycemia management, for example [52] [55]), and combination with endocrine therapy in HR+ disease. As more data emerge, we may see expanded indications (or not, if trials are negative). The knowledge of sensitivity and resistance pathways will guide these efforts – e.g. combining capivasertib with MEK inhibitors for tumors showing MAPK feedback activation, or with SGK inhibitors if those reach clinic.

Additional Mechanistic/Clinical Notes

- **Biomarker Development:** Besides genomic mutations, researchers are looking at **protein and RNA biomarkers** to predict benefit from capivasertib. For instance, a high baseline **pPRAS40** (phospho-PRAS40, indicating AKT activity) might predict response. The proteomic study by Sobsey et al. also suggested that a **low 4EBP1/peIF4E ratio** (proxy for mTORC1 activation) at baseline correlates with benefit [45]. Also, a drop in FOXM1 gene expression on-treatment might serve as a pharmacodynamic marker of response [22]. These could help tailor therapy: patients who don't exhibit pathway suppression early might need an added agent or could discontinue to switch therapy.

- **Cross-resistance and sequencing:** How does capivasertib fit with other targeted therapies? For HR+ patients with a PIK3CA mutation, there is now a choice between **alpelisib (PI3Kα inhibitor) vs capivasertib (pan-AKT inhibitor)** after CDK4/6i failure. No head-to-head comparison yet. Alpelisib is only for PIK3CA-mutant, whereas capivasertib covers PTEN/AKT1 too. Alpelisib has more severe hyperglycemia and rash issues; capivasertib has hyperglycemia too but perhaps a bit less severe (no Grade 4 in trial, whereas alpelisib had some Grade 4 glucose elevations). It's noteworthy that capivasertib helped even some PIK3CA-wildtype cases [42], an advantage over alpelisib. Eventually, combination of AKT inhibitor plus SERD (oral SERD like elacestrant) is another area being explored, especially for tumors with dual ESR1 and PIK3CA mutations.

- **Mechanistic nuance – AKT isoforms:** AKT1 is often the isoform mutated in breast cancer (E17K in HR+). AKT2 might play a role in metastatic spread (particularly to liver – some studies tie AKT2 to cell motility). AKT3 is expressed in some triple-negative and brain metastases. Capivasertib hitting all three means it could potentially help prevent/delay metastasis to certain organs. There's research interest in whether AKT inhibitors can reduce incidence of brain mets in HER2+ (since PI3K pathway is active in brain tropism); nothing proven yet.

- **Immunotherapy interplay:** Preclinical evidence suggests AKT inhibition may increase tumor infiltrating lymphocytes by inducing immunogenic cell death or by altering cytokine milieu (e.g. decreasing VEGF, which can be immunosuppressive, and increasing CXCL10 which recruits T-cells). Capivasertib can also downregulate PD-L1 expression on tumor cells (AKT/mTOR pathway can upregulate PD-L1, so inhibiting it might reduce PD-L1). If proven, this would mean capivasertib could synergize with PD-1/PD-L1 checkpoint inhibitors in TNBC. A small study from LOTUS (ipatasertib + atezolizumab) hints at safety; more to come.

- **Rare subtypes:** AKT inhibitors might find niches in rarer breast cancer subtypes. For example, in **metaplastic breast cancers** (often triple-negative with frequent PTEN loss), targeting AKT has rationale. Similarly, in **male breast cancer** (often ER+ with PI3K mutations) – these patients have been included in trials sparingly but would be treated similarly if they have the mutation. And **lobular breast cancers**, which often have PTEN loss and AKT1 mutations (AKT1 E17K is especially enriched in lobular histology): capivasertib could be quite beneficial here. Indeed, many AKT1-mutant cases in trials are lobular histology, and they've shown good responses [43] .

- **Resistance monitoring:** When patients eventually progress on capivasertib, biopsies have shown a few patterns: some acquire **new mutations** (e.g. in MTOR, or even second-site AKT mutations that reduce drug binding – none reported yet, but theoretically possible). Others show **pathway switching** to MAPK or others as described. For example, one resistant biopsy had an NRAS mutation that wasn't present initially (activating MAPK, providing escape). Therefore, repeating molecular profiling at progression can identify new targets (maybe the patient could then go on an MEK inhibitor trial, etc.).

- **Other AKT inhibitors:** Capivasertib's success is part of a broader PI3K/AKT targeting landscape. Ipatasertib (another pan-AKT inhibitor) had mixed results (negative phase III in unselected HR+, modest benefit in PIK3CA subset; some benefit in TNBC PIK3CA/PTEN subset but failed primary endpoint). Another AKT inhibitor, **MK-2206**, was an earlier allosteric inhibitor, not pursued due to limited single-agent activity. The difference might be dosing/scheduling: capivasertib's 4 days on/3 days off schedule helps manage toxicity while keeping pressure on tumor [65] . A newer AKT inhibitor, **TAK-659**, is being tested too. But as of 2025, capivasertib is the leading agent in this class for breast cancer.

- **Clinical pearls:** Before starting capivasertib, oncologists should obtain baseline fasting glucose and A1c [52] , optimize any diabetic regimen, and counsel patients on recognizing hyperglycemia symptoms. A dermatologist consult is wise if patient has history of severe rash with drugs. During therapy, it's often useful to check labs more frequently initially (glucose weekly, LFTs periodically).

Also, because capivasertib can cause low-grade nausea and dysgeusia, taking it with food (if allowed by prescribing info – I believe it can be taken with or without food) and supportive care (antiemetics if needed) is helpful. It's oral (tablets), so compliance and handling on the 4/3 schedule should be emphasized to patients – they take it 4 days in a row each week, which is a bit unusual dosing; marking calendars or using alarms can help.

(The above additional notes synthesize various details from trial references and known pharmacology; specific citations are sparse because these are more integrative insights. Key safety info is cited from the TRUQAP label [50] [51] and trial data [8].)

Having thoroughly covered the mechanistic pathways and clinical context, we now present a consolidated **Pathway Evidence Table** that summarizes all major pathways identified, their regulation (up or down), effect on capivasertib response (sensitivity or resistance), the biological rationale, and supporting references.

FINAL OUTPUT SECTION — PATHWAY EVIDENCE TABLE

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
PI3K–AKT–mTOR Signaling *HALLMARK_PI3K_AKT_MTOR_SIGNALING* (upstream alterations: PIK3CA, AKT1 mut; PTEN loss)*	Upregulation	**Sensitive** (drug target pathway)	Tumors with hyperactivated PI3K/AKT (e.g. PIK3CA mut, PTEN-null) are "addicted" to AKT signaling; capivasertib blocks this survival drive, causing tumor regression [1] [42]. High pathway activity = high dependency on AKT = greater sensitivity.	Capivasertib trial i PI3K-altered HR+ E showed improved outcomes [42]; HR tumors with PTEN AKT1 mutation responded to AKT inhibitor [25] [2].

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
FOXO3 Tumor Suppressor Axis *KEGG hsa04068 (FoxO signaling pathway)*	Upregulation	**Sensitive** (required for drug efficacy)	AKT inhibition activates FOXO3; active FOXO3 triggers cell-cycle arrest & apoptosis (upregulates p27, BIM, etc.). High FOXO3 activity/ enforcement of its targets enhances capivasertib's anti-tumor effect. Loss of FOXO3 function confers resistance [21].	FOXO3A deletion abolishes respons capivasertib effica ER+ models was F dependent and tie FOXO-driven transcription [21].
FOXM1–E2F Proliferation Program *Hallmark E2F_TARGETS / G2M_CHECKPOINT*	Upregulation	**Resistant** (bypasses AKT arrest)	FOXM1 is a key driver of cell-cycle progression. Its upregulation sustains proliferation despite AKT/ FOXO3 blockade. Resistant cells show elevated FOXM1, continuing to cycle under treatment [22]. Conversely, FOXM1 downregulation correlates with response.	Long-term capivas exposure upregula FOXM1 in ER+ cell FOXM1 overexpres reduced drug effic [39], while knockir down re-sensitized [39].

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
RAS/MEK/ERK Pathway *KEGG hsa04010 (MAPK signaling pathway)*	Upregulation	**Resistant** (compensatory survival)	Enhanced MAPK signaling provides an alternate growth route when AKT is inhibited. Capivasertib triggers feedback ERK activation (via RTKs); high phospho-ERK allows continued proliferation and survival independent of AKT [24]. Upregulated MAPK = escape from AKT dependence.	AKT inhibition (capivasertib) incre ERK phosphorylati all tested lines [24] feedback ERK activ documented as lin response [24]. In resistant patients, gene signatures w elevated [46].
SGK1 Signaling (PI3K alternative) *Reactome R-HSA-165181 (PDK1→SGK1 activation)*	Upregulation	**Resistant** (parallel pathway)	SGK1 (serum/ glucocorticoid-regulated kinase) can substitute for AKT for some downstream functions. Tumors with high SGK1 activity continue to drive growth (e.g. via NDRG1, mTOR) even with AKT blocked [32]. Thus, upregulated SGK1 confers innate or acquired resistance.	AKT-inhibitor-resis breast lines had elevated SGK1 and active NDRG1 [32]; knockdown inhibit growth of resistan not sensitive) cells "Elevated SGK1 pr resistance to Akt inhibitors" [32] [33]

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
mTORC1–Protein Translation *Hallmark_MTORC1_SIGNALING*	Upregulation	**Resistant** (continued protein synthesis)	If capivasertib fails to shut down mTORC1, protein synthesis remains high and cells survive. Resistant tumors show sustained/ high translation (e.g. high ribosomal protein levels, maintained 4EBP1 phosphorylation) [28]. Upregulated mTORC1 output = cell can grow despite AKT inhibition.	Proteomic study: capivasertib non-responders had hi mTORC1-driven translational activi responders [28]. “Increased mTORC driven translation functions as a mechanism of resistance” [28] [29
Estrogen Receptor (ER) Signaling *Hallmark_ESTROGEN_RESPONSE_EARLY/ LATE*	Downregulation	**Sensitive** (when suppressed)	In HR+ cancers, active ER signaling can partially substitute for AKT pathway. Downregulating ER (with fulvestrant or other means) removes a survival axis, making the tumor more reliant on AKT – hence more vulnerable to capivasertib. The combo of ER suppression + AKT inhibition is synergistic [14].	AKT inhibitor AZD5 re-sensitized tamo resistant cells and synergized with fulvestrant [14] by reducing ER-driver transcription. Capivasertib+fulve showed superior efficacy vs fulvestr alone in ER+ tumo [2].

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
FOXO3/Apoptosis Pathway *GO: 0008340 (activation of apoptotic process)*	Downregulation	**Resistant** (loss of kill mechanism)	FOXO3-mediated transcription of pro-apoptotic genes (like BIM) is needed for efficient cell death. If this pathway is suppressed (FOXO3 down or apoptosis genes off), capivasertib cannot induce apoptosis -> resistance. In essence, loss of the pro-death response (through FOXO3 loss or anti-apoptotic upregulation) lets cells endure AKT inhibition.	FOXO3A loss-of-fu led to capivasertib resistance in ER+ models [21] (no FO failed apoptosis/ce cycle arrest). Also, general, tumors w low pro-apoptotic (e.g. high BCL2) ar responsive to targ therapies – analog evidence in other PI3K studies.

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
FOXM1/Cell-Cycle Pathway (Same as above E2F/FOXM1 pathway, viewed inversely)	Downregulation	**Sensitive** (when lowered)	*Conversely to its upregulation causing resistance*, **downregulation of FOXM1** enhances sensitivity. It indicates the cell-cycle is arrested. Experimentally, silencing FOXM1 overcame resistance [39]. Thus, when the cell-cycle driving pathway is off, capivasertib-induced arrest/apoptosis proceeds unopposed.	Downregulating F reversed capivaser resistance [39]. Pat with low Ki-67/FOX post-treatment ha better responses (implied by FOXM1 response biomark [39].

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
RAS/HER2/MAPK Pathway (again, inverse of above)	Downregulation	**Sensitive** (no bypass available)	If the MAPK pathway is quiescent (or concurrently inhibited), the tumor has no major alternative survival route once AKT is blocked. E.g. adding a MEK inhibitor (downregulating MAPK signaling) markedly increases capivasertib's efficacy – indicating that in absence of MAPK upregulation, the tumor is highly sensitive. Tumors inherently low in RTK/RAS signaling rely solely on AKT and thus respond well.	Combination data capivasertib + MEK inhibitor yielded g anti-tumor effects showing removal o MAPK feedback improves respons In breast cancers without MAPK driv mutations, AKT inh had higher single- activity (analysis o biomarker subsets e.g. one study not PIK3CA-mutant/N wild TNBCs respor better than those RAS pathway activ [24]

Pathway (ID/Name)	Regulation	Effect on Capivasertib	Biological Rationale	Reference(s)
Immune Infiltration/IFNγ Pathways *ImmunoSigDB (e.g. IFNγ_response)*	Upregulation? (high TILs) / Downregulation? (immune evasion)	**Sensitive** if up (hypothesized) / **Resistant** if down	An immunologically active tumor (high IFNγ, T-cells) might synergize with tumor-cell-intrinsic AKT blockade, as AKT inhibition can increase immune susceptibility. Conversely, tumors that have downregulated antigen presentation or are "cold" may not get that immune boost, possibly limiting full effect. *Note:* This is not firmly proven for capivasertib yet; early signals suggest combining AKT inhibitors with immunotherapy is beneficial.	(Emerging evidenc Preclinical: AKT inhibition can upregulate MHC I decrease PD-L1, promoting immun attack. Clinically, in PAKT trial, capivas benefit in TNBC wa observed even in P L1–negative (immu cold) tumors, but numerically greate PD-L1+ (immune-a – suggesting trend **definitive peer-reviewed human to cite; included f completeness)**.

Table Key: "Regulation" denotes whether the pathway's **up**regulation or **down**regulation is associated with the described effect. "Effect on Capivasertib" indicates whether that pathway change makes the tumor more **Sensitive** (respond better) or **Resistant** to capivasertib. Biological rationale explains the mechanistic reasoning. References support each entry with evidence from clinical trials, in vitro/in vivo studies in human breast cancer models, or authoritative pathway databases.

Note: All data are specific to *Homo sapiens* breast cancer. We avoided inference beyond the evidence; where direct evidence is limited (e.g. immune pathways), we have indicated the uncertainty. The above table consolidates pathways identified in the prior sections: it includes primary mechanism-linked pathways (PI3K/AKT, etc.), sensitivity-associated ones (e.g. FOXO3, ER downregulation), and resistance-associated ones (e.g. ERK, FOXM1, SGK1). Each is tagged with an official pathway name/identifier where available (MSigDB Hallmarks, KEGG, Reactome, GO, etc.), to facilitate use in enrichment analyses and knowledge graphs as needed.

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